

# **HVAC System Optimization Report**

Ottawa Commercial Building Case Study

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Realistic TMYx Weather • Part-Load Curves • Monte Carlo Analysis

# Executive Summary

## Baseline Performance

Annual Energy Consumption : 636,413 kWh  
Annual Operating Cost : \$76,369.55

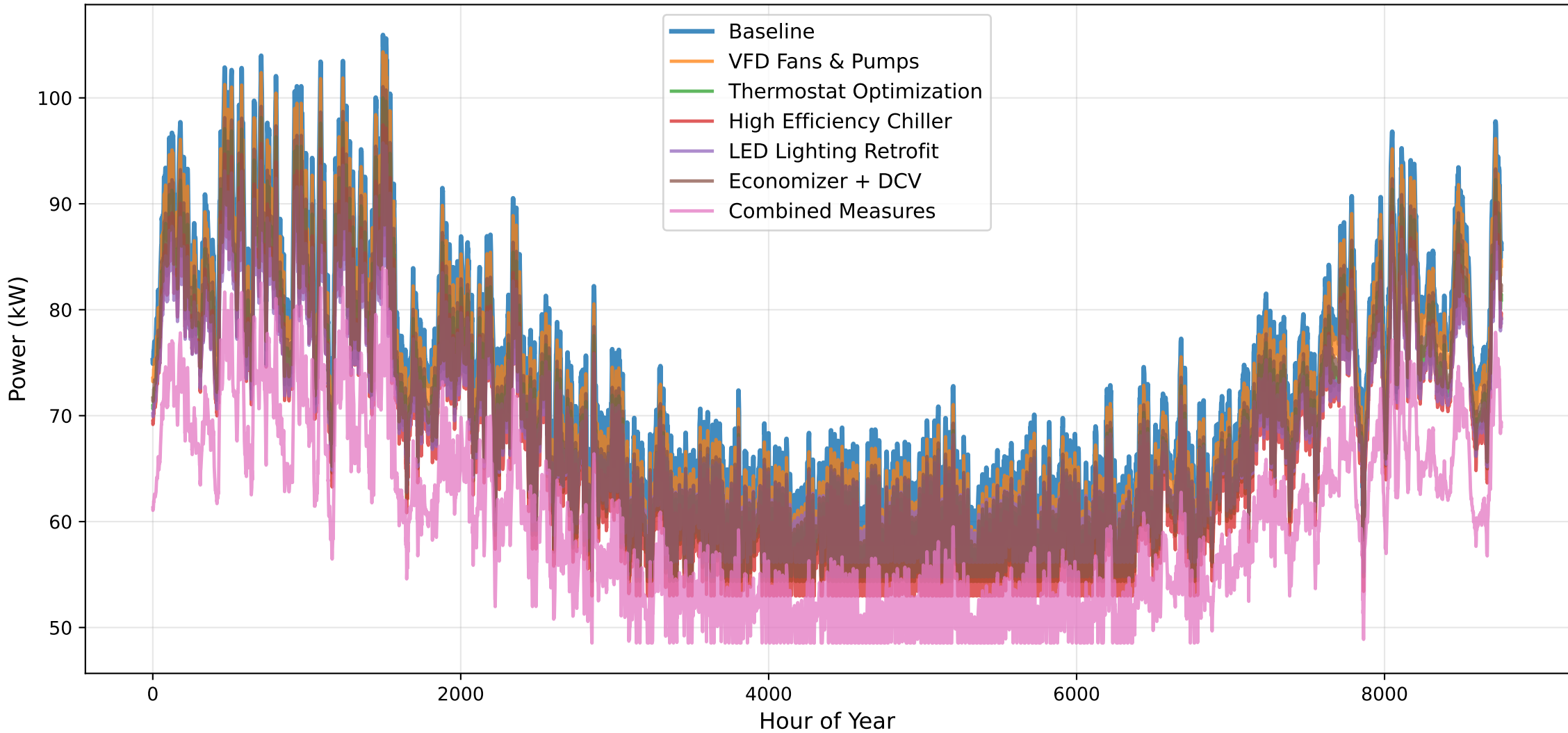
## Best Scenario – Combined Measures

Annual Energy : 520,613 kWh  
Energy Reduction : 115,800 kWh (18.2%)  
Cost Savings : \$13,896.02  
Payback Period : 3.6 years  
CO<sub>2</sub> Reduction : 57.9 tonnes

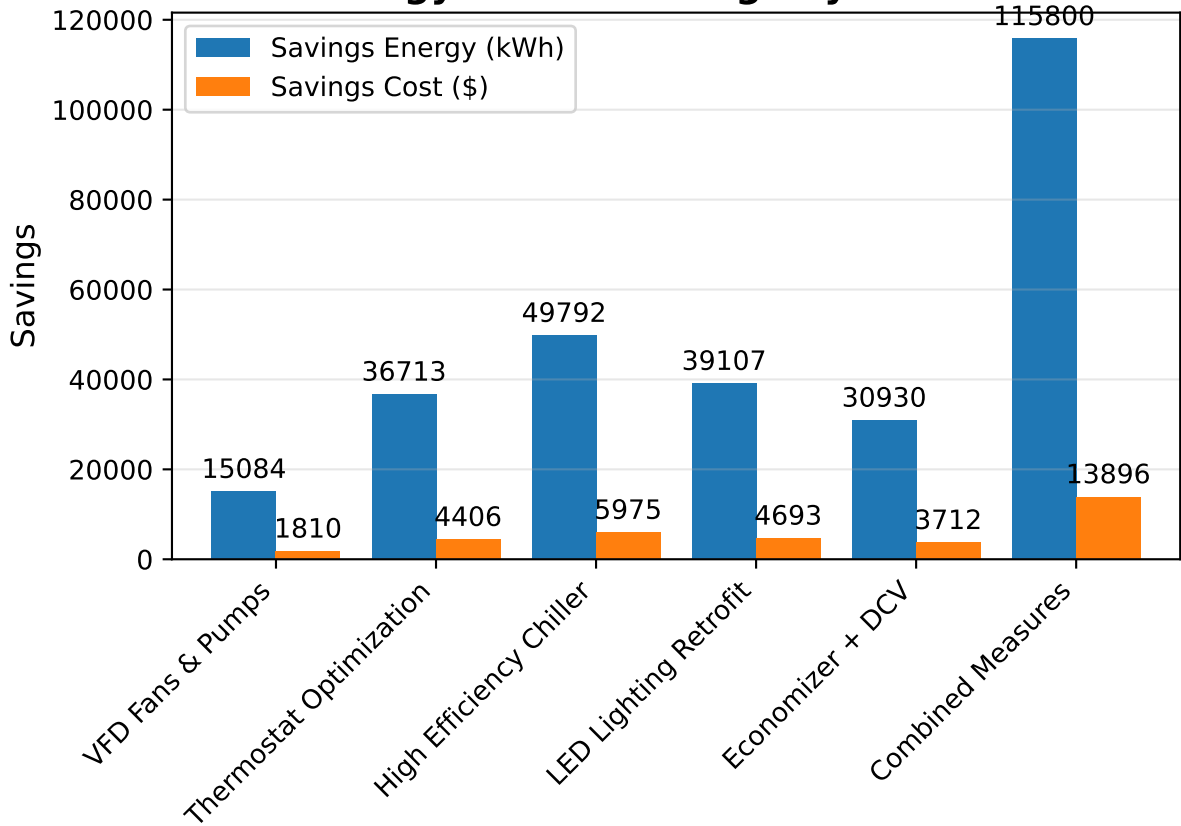
## Parametric Optimization (SciPy)

Additional Annual Savings : \$-8,969  
Minimum Annual Cost Found : \$53,504.14

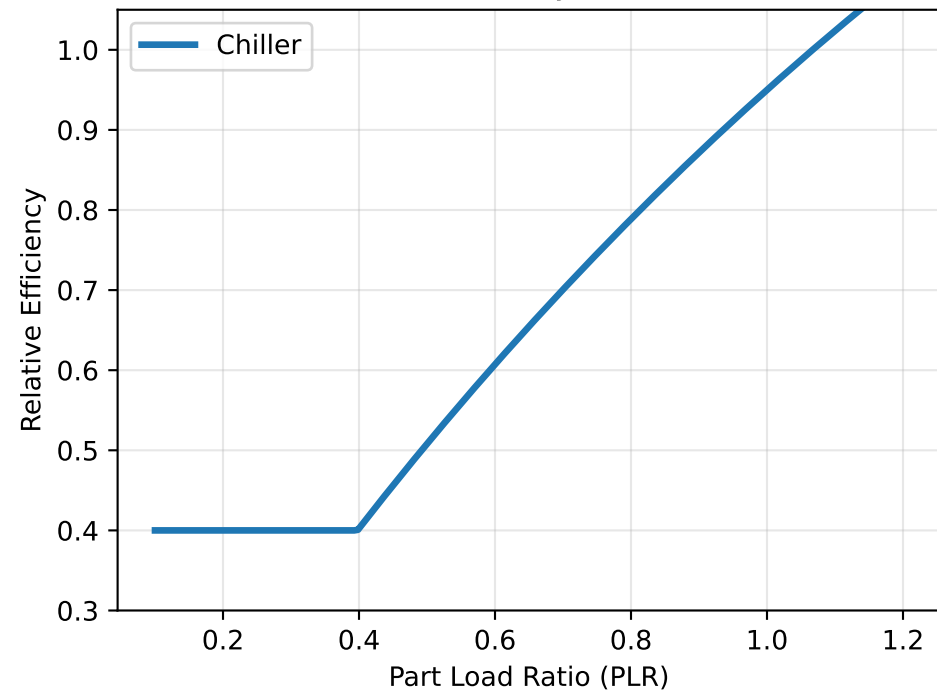
# Hourly HVAC Energy Consumption Over One Year



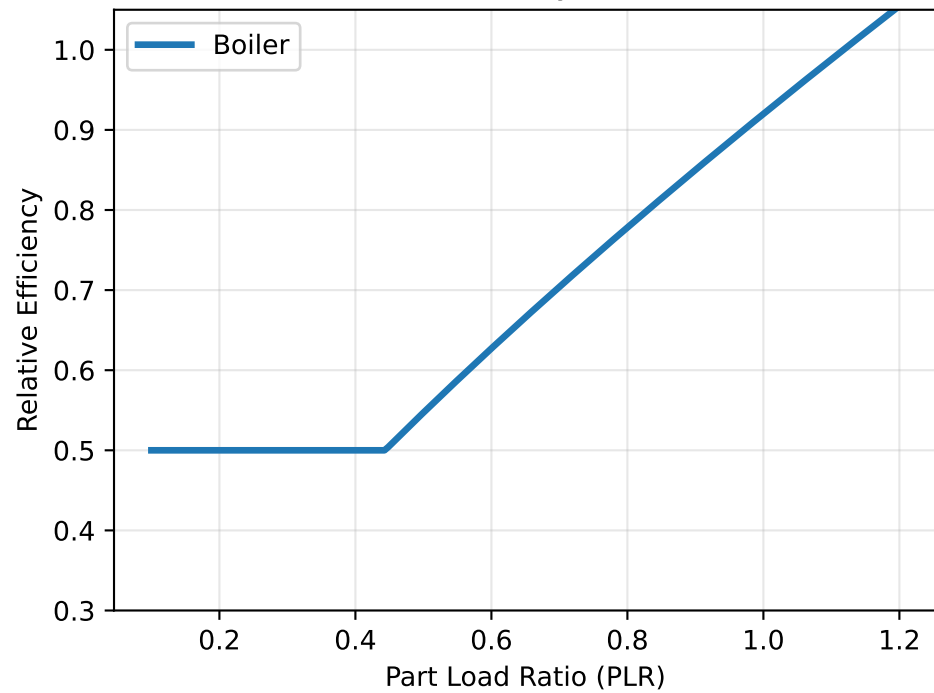
# Energy & Cost Savings by Scenario



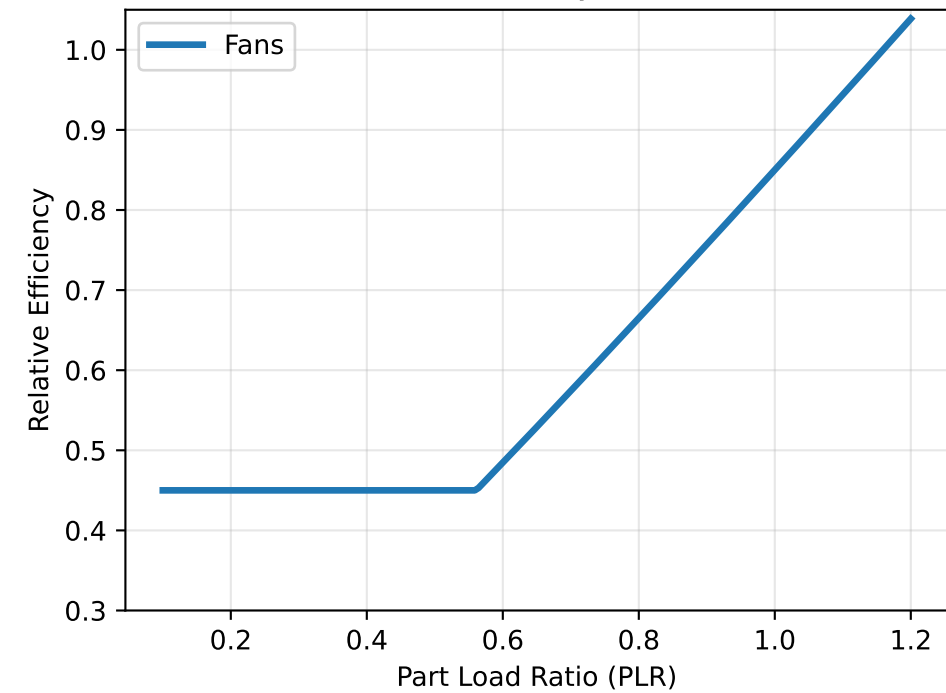
Part-Load Efficiency Curve: Chiller



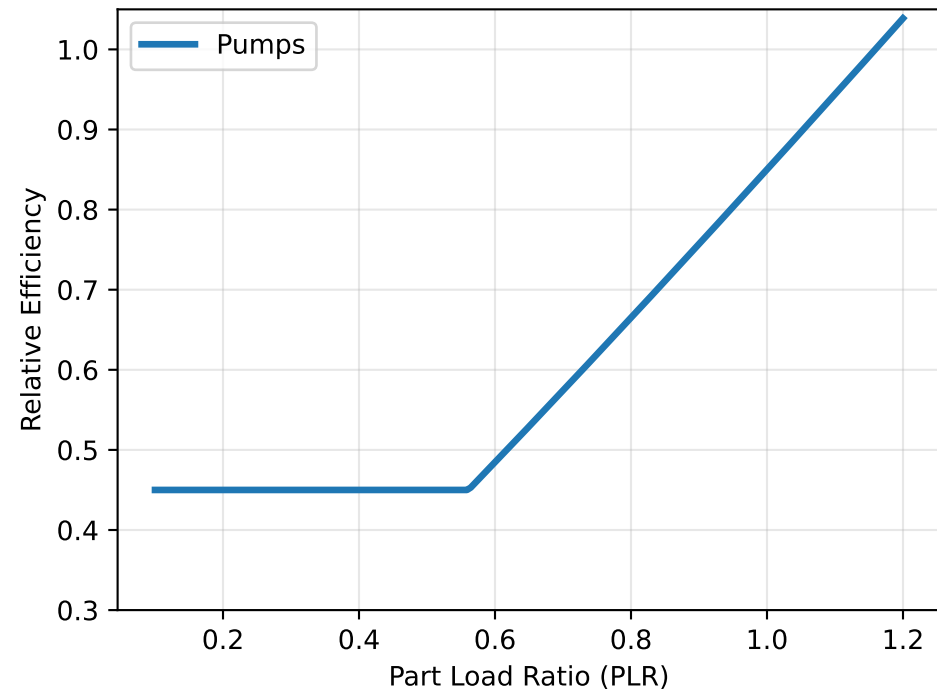
Part-Load Efficiency Curve: Boiler



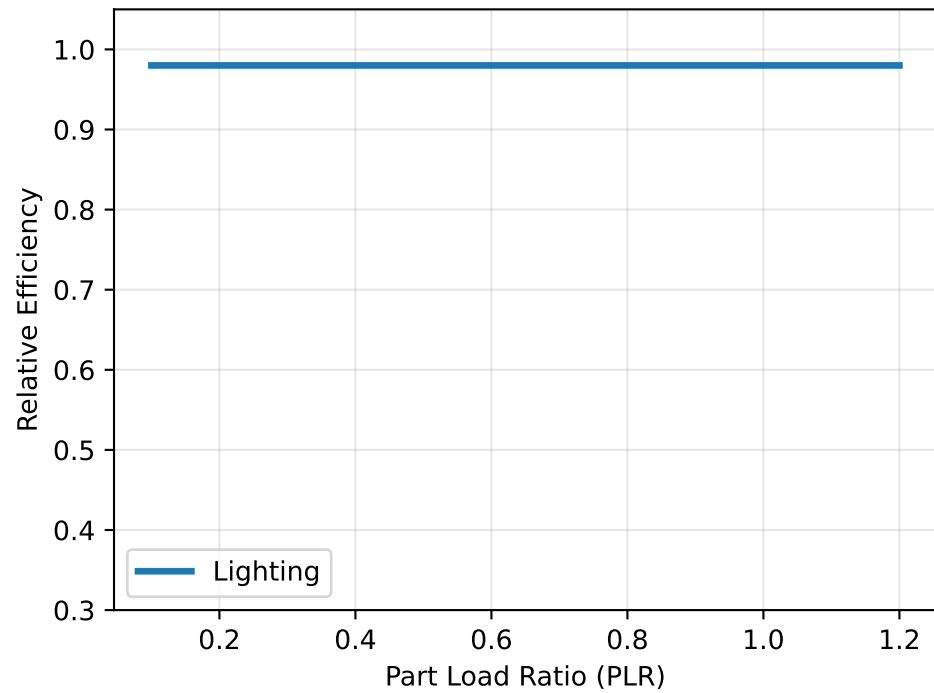
Part-Load Efficiency Curve: Fans



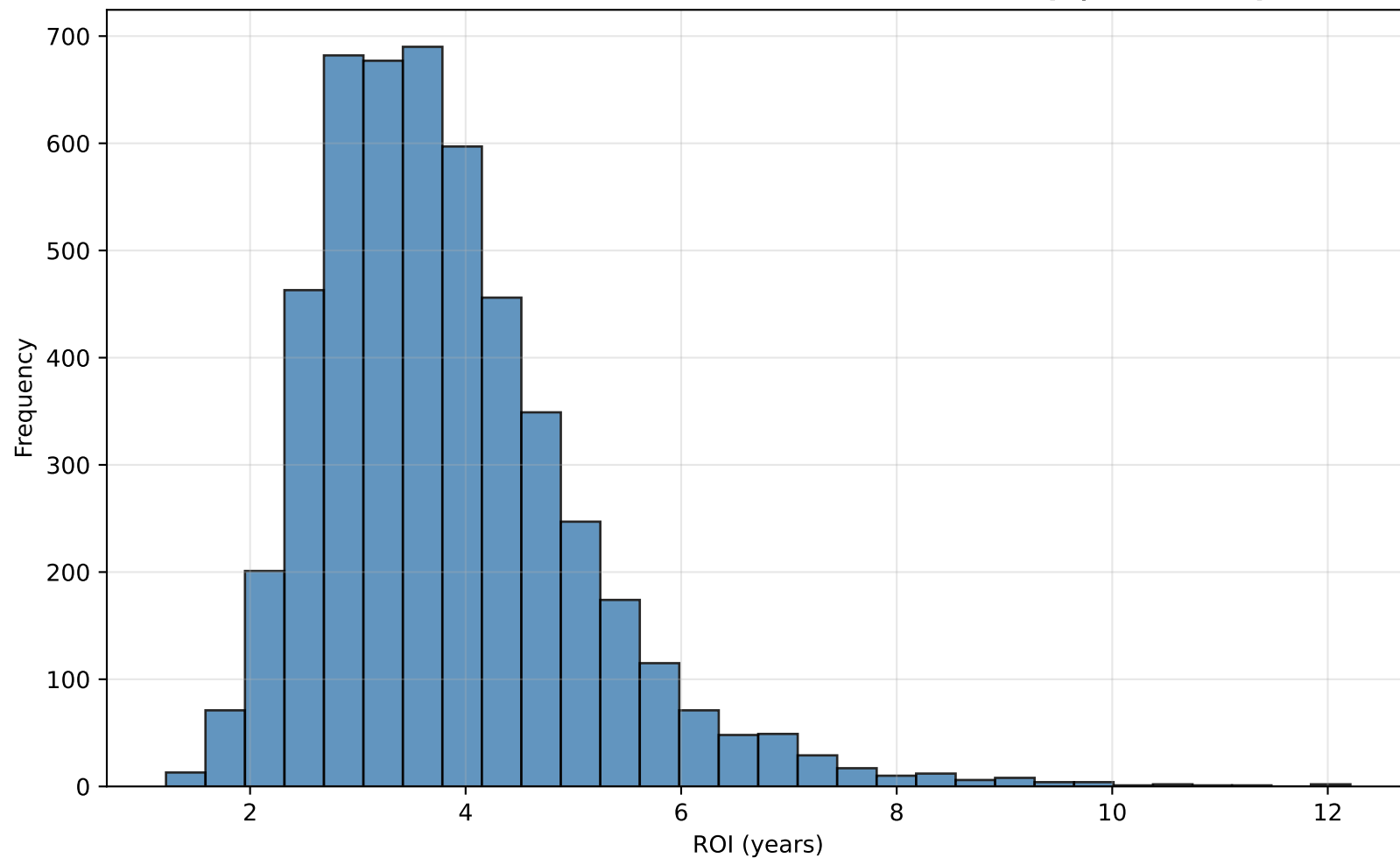
Part-Load Efficiency Curve: Pumps



Part-Load Efficiency Curve: Lighting



**Monte Carlo Simulation — ROI Distribution (5,000 runs)**



## Optimal Configuration from Parametric Optimization

Recommended Upgraded Component Ratings:

|          |   |         |                    |
|----------|---|---------|--------------------|
| Chiller  | : | 22.5 kW | ( 25.0% reduction) |
| Boiler   | : | 12.0 kW | ( 20.0% reduction) |
| Fans     | : | 3.2 kW  | ( 36.0% reduction) |
| Pumps    | : | 3.5 kW  | ( 30.0% reduction) |
| Lighting | : | 3.2 kW  | ( 60.0% reduction) |